Image Sampling and Quantization

Rama Moorthy H

Dept of BCA, NIPE

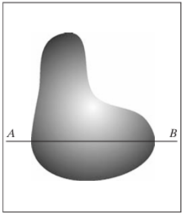
Overview

Goal

Convert sensed data into a **digital image**.

Key Steps

- **Sampling** – Digitizing spatial coordinates (x, y).
- **Quantization** – Digitizing intensity (amplitude) values.



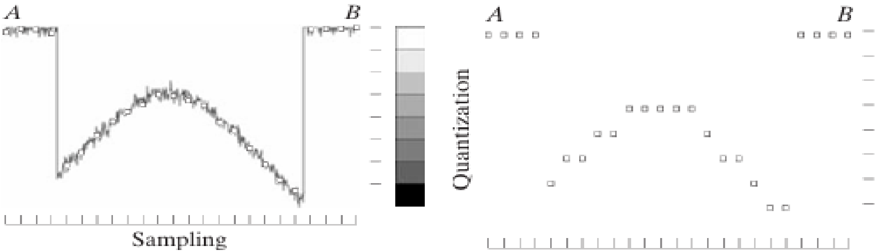
Continuous to Digital Image

Sampling

- Select equally spaced spatial points.
- Determines image resolution.

Quantization

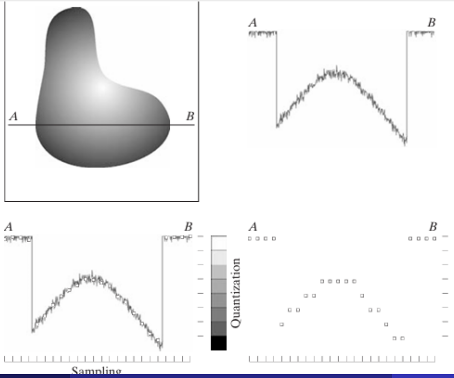
- Map continuous intensity values to discrete levels.
- Affected by noise in the signal.



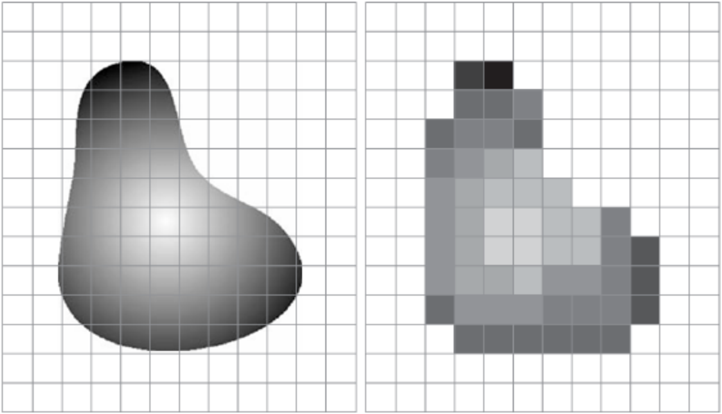
1D Example: From Continuous to Digital

Steps

1. Take a scan line from the continuous image.
2. Apply **sampling** – choose spatial positions.
3. Apply **quantization** – assign intensity levels.
4. Repeat line-by-line to build a 2D image.



Result



Practical Considerations

Sensor Configurations

- **Single Sensor + Mechanical Motion**: Fine sampling possible; limited by optics.
- **Sensor Strip**: One direction limited by sensor count, other by motion.
- **Sensor Array**: No motion; resolution fixed by number of sensors in both directions.

Image Quality Factors

- Number of samples (spatial resolution).
- Number of intensity levels (quantization).
- Image content and noise level.

Representing Digital Images

Idea

A continuous image $f(s, t)$ is converted into a **digital image** by:

- **Sampling** $\rightarrow$ Produces a 2D array of $M \times N$ elements.
- Discrete coordinates: $x = 0, 1, \ldots, M - 1$ and $y = 0, 1, \ldots, N - 1$.
- Each element $f(x, y)$ stores intensity at integer coordinates.

Spatial Domain

The (x, y) coordinates define the **spatial domain**, and are called spatial variables.

Three Basic Representations of $f(x, y)$

1. Surface Plot

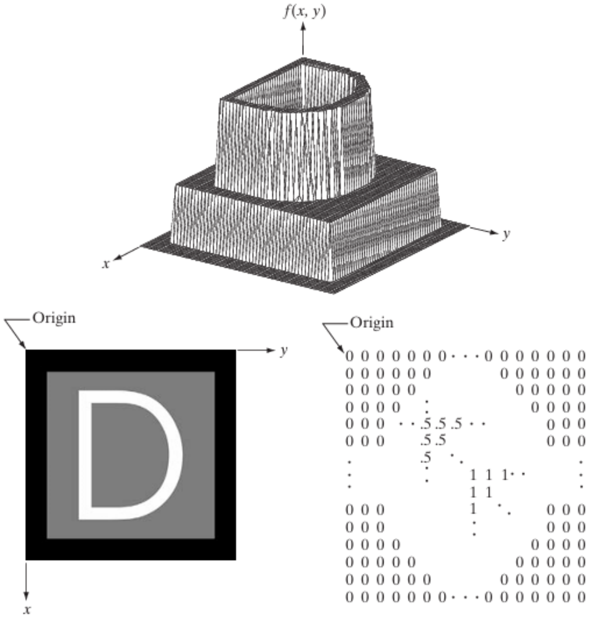
- 3D plot: (x, y, z) where $z = f(x, y)$.
- Useful for simple grayscale images.
- Hard to interpret for complex images.

2. Visual Intensity Array

- Displayed as an image (monitor, print).
- Pixel brightness proportional to $f(x, y)$.
- Example: $0 \rightarrow$ black, $0.5 \rightarrow$ gray, $1 \rightarrow$ white.

3. Numerical Array

- $M \times N$ matrix of numbers.
- Useful for algorithm development.
- Example: $600 \times 600 \rightarrow 360{,}000$ values.


f(x, y)
x
y
Origin
y
x
Origin
0 0 0 0 0 0 0 · · · 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0
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Mathematical Representation

Matrix Form

$$f(x,y) = \begin{bmatrix} f(0,0) & f(0,1) & \dots & f(0,N-1) \\ f(1,0) & f(1,1) & \dots & f(1,N-1) \\ \vdots & \vdots & \ddots & \vdots \\ f(M-1,0) & f(M-1,1) & \dots & f(M-1,N-1) \end{bmatrix}$$

Terminology

- Each element = **image element**, **picture element**, **pixel**, or **pel**.
- Alternative notation: $a_{ij} = f(i,j)$.

$$\mathbf{A} = \begin{bmatrix} a_{0,0} & a_{0,1} & \cdots & a_{0,N-1} \\ a_{1,0} & a_{1,1} & \cdots & a_{1,N-1} \\ \vdots & \vdots & & \vdots \\ a_{M-1,0} & a_{M-1,1} & \cdots & a_{M-1,N-1} \end{bmatrix}$$